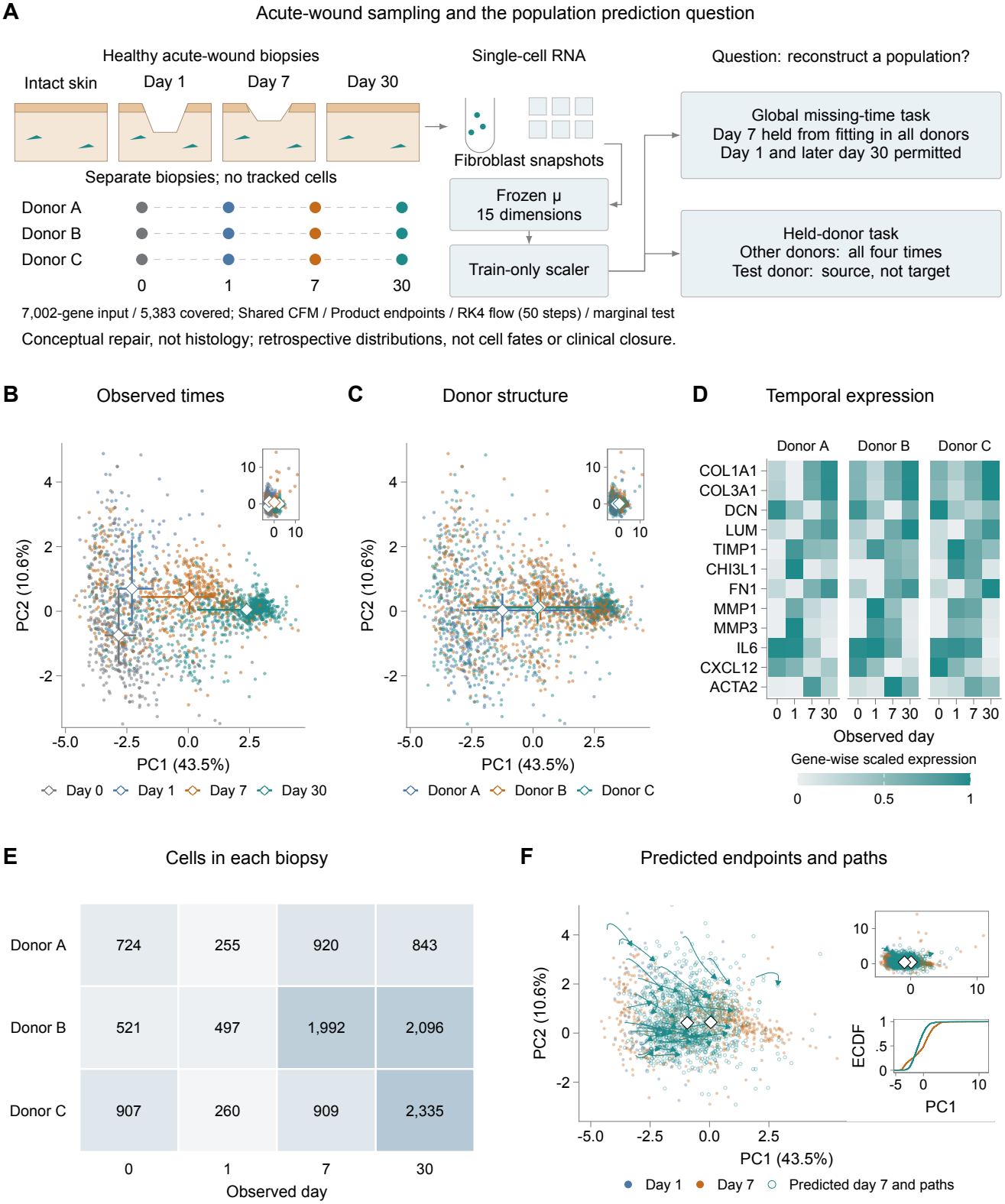

A
Acute-wound sampling and the population prediction question
Healthy acute-wound biopsies
Intact skin
Day 1
Day 7
Day 30
Separate biopsies; no tracked cells
Donor A
Donor B
Donor C
0
1
7
30
Single-cell RNA
Fibroblast snapshots
Frozen μ
15 dimensions
Train-only scaler
Question: reconstruct a population?
Global missing-time task
Day 7 held from fitting in all donors
Day 1 and later day 30 permitted
Held-donor task
Other donors: all four times
Test donor: source, not target
7,002-gene input / 5,383 covered; Shared CFM / Product endpoints / RK4 flow (50 steps) / marginal test
Conceptual repair, not histology; retrospective distributions, not cell fates or clinical closure.
B
Observed times
PC2 (10.6%)
PC1 (43.5%)
Day 0
Day 1
Day 7
Day 30
C
Donor structure
PC2 (10.6%)
PC1 (43.5%)
Donor A
Donor B
Donor C
D
Temporal expression
Donor A
Donor B
Donor C
COL1A1
COL3A1
DCN
LUM
TIMP1
CHI3L1
FN1
MMP1
MMP3
IL6
CXCL12
ACTA2
0 1 7 30
Observed day
Gene-wise scaled expression
0
0.5
1
E
Cells in each biopsy
Donor A 724 255 920 843
Donor B 521 497 1,992 2,096
Donor C 907 260 909 2,335
0
1
7
30
Observed day
F
Predicted endpoints and paths
PC2 (10.6%)
PC1 (43.5%)
ECDF
PC1
Day 1
Day 7
Predicted day 7 and paths